ENCE 3210 – Microprocessors 1

Lab 4

February 27th, 2026

Dead Line: March 6th, 2026

****

Directions: Create a GitHub repository for this assignment and upload your answers there. Provide the GitHub link on the respective Canvas assignment.

****

Q1 – The code below was given to you by a firmware engineering company that wants you to improve it. The main goal is to simulate in software a 14-bit SAR and test the conversion for the following analog voltage levels: 0.42V, 0.83V, 1.65V, and 2.752V with a reference voltage of 3V.

```
#define bitsize 12

float Vref = 3.6;
float Vin = 3.3;
float thresh;
float quantized = 0;
int count;
int bitval;
int bits[bitsize];

void main (void){

    Vref /= 2;
    thresh = Vref;

    for(count=0; count<bitsize; count++){
        Vref /= 2;
        if (Vin >= thresh){
            bitval = 1;
            thresh += Vref;}
        else {
            bitval = 0;
            thresh -= Vref;}

        bits[count] = bitval;
        quantized += 2*Vref*bitval;}

    while(1);
}
```

Q2 – You are responsible to design a solar charge controller code using your embedded platform with the specifications below:

a) The system has a battery, a solar panel, a microcontroller, and a controllable switching element (such as a power multiplexer or analog switch IC).
b) If the voltage output of the solar panel is greater than 2.5V, then the microcontroller power is supplied from the solar panel.
c) If the voltage level of the solar panel is lower than 2.5V, then the battery will supply power to the microcontroller.
d) If the voltage level of the battery drops below the voltage of the solar panel, then the switching element connects the battery to the solar panel.
e) Provide a block diagram of this system.
f) For software purposes, since we don't have a access to a controllable switching element, use the OLED display to indicate how the power is routed in the system.

Q3 – The solar charge controller specifications in Q2 have a major issue, the system never has the chance to charge the battery fully.

a) Improve the code from Q2 to allow charging the battery fully before switching the solar panels back to the microcontroller.
   [Note: if for some reason your solution in Q2 addresses this issue, disregard this question]

Q4 – Design a fan controller using your embedded platform with the specifications below:

a) The system has the following components: an analog temperature sensor, a fan, and 2 user buttons.
b) The temperature sensor will be monitored using the ADC module. You will measure 100 samples over a 5 second period and calculate the average temperature to find a reliable value.
c) Then generate a PWM signal to change the speed of the fan according to the calculated temperature value. Set the PWM frequency as 250 Hz.
d) Use button 1 to turn on or off the fan.
e) Use button 2 to change the sensitivity of the fan speed with respect to temperature. Use three sensitivity levels such that the lower the sensitivity the lower the speed of the fan.
f) Since the complexity of your code starts to increase, provide a code diagram of your solution.
g) For software purposes, since we don't have access to a motor/fan, use an LED that is connected to a pin with PWM capabilities. The brightness of the LED will simulate the speed of the fan.